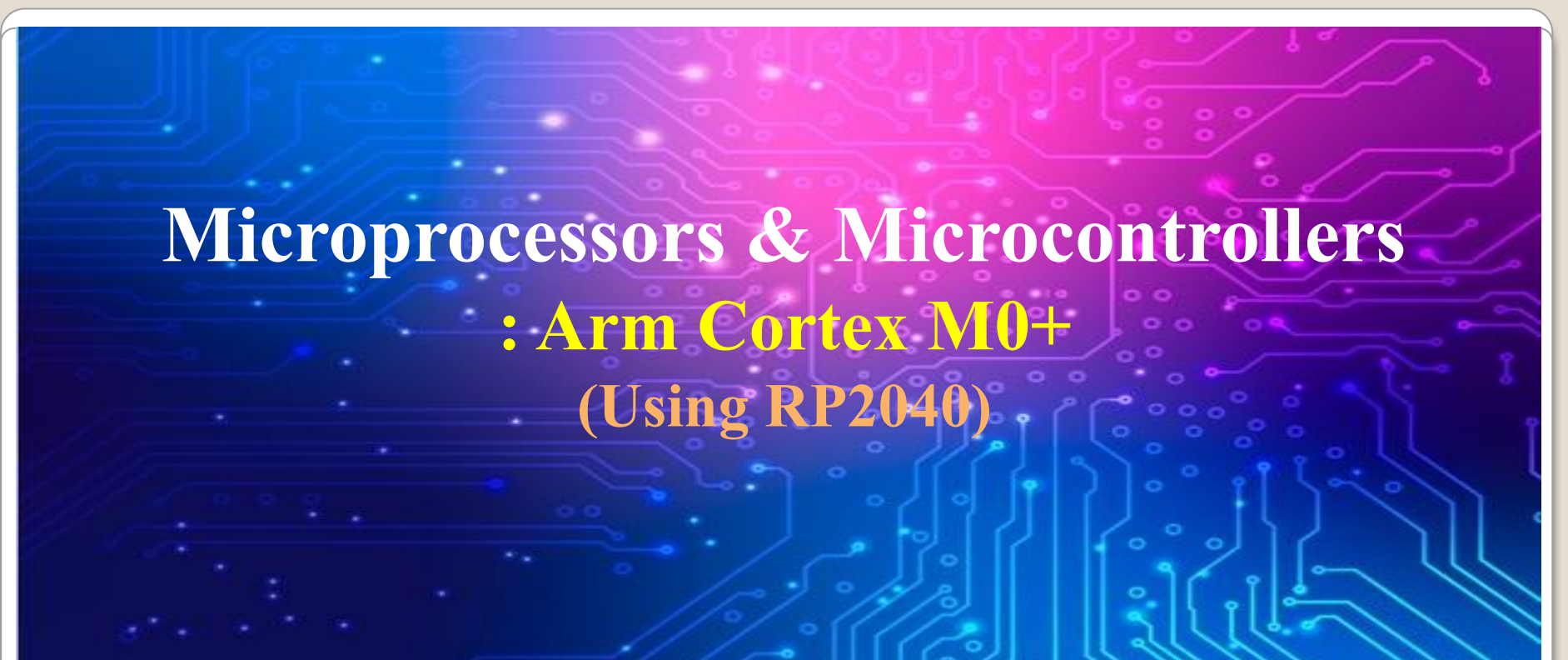




Microprocessors & Microcontrollers

: Arm Cortex M0+

(Using RP2040)

Lecture 1.2.5

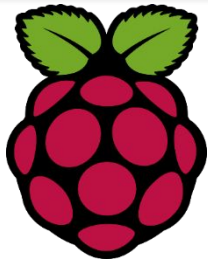
Introduction to Raspberry Pi Family

Mouli Sankaran

Lecture 1.2.5: Focus

- Introduction to Raspberry Pi Family
 - Introduction
 - Raspberry Pi Pico
 - Raspberry Pi Pico W

Raspberry Pi Family (Introduction)



Raspberry Pi Logo

The **Raspberry Pi Foundation** is a **British charity and company** founded in **2009**, to promote the **study of basic computer science** in **schools**, and is responsible for developing the **Raspberry Pi single-board computers**.

A few Boards of Raspberry Pi family

Raspberry Pi Zero
Released in Nov 2015



Raspberry Pi 3 Model B+
Released in 2016



Raspberry Pi 4 B
Released in Jun 2019



- **Raspberry Pi Pico** was released in **January 2021** with a retail price of **\$4**.
- It was Raspberry Pi's first board, based on a single **ARM microcontroller** chip designed in the UK.
- It is programmable in MicroPython, CircuitPython, and C/C++.
- It is designed for physical computing, similar in concept to an Arduino.

Raspberry Pi Pico (RP2040)

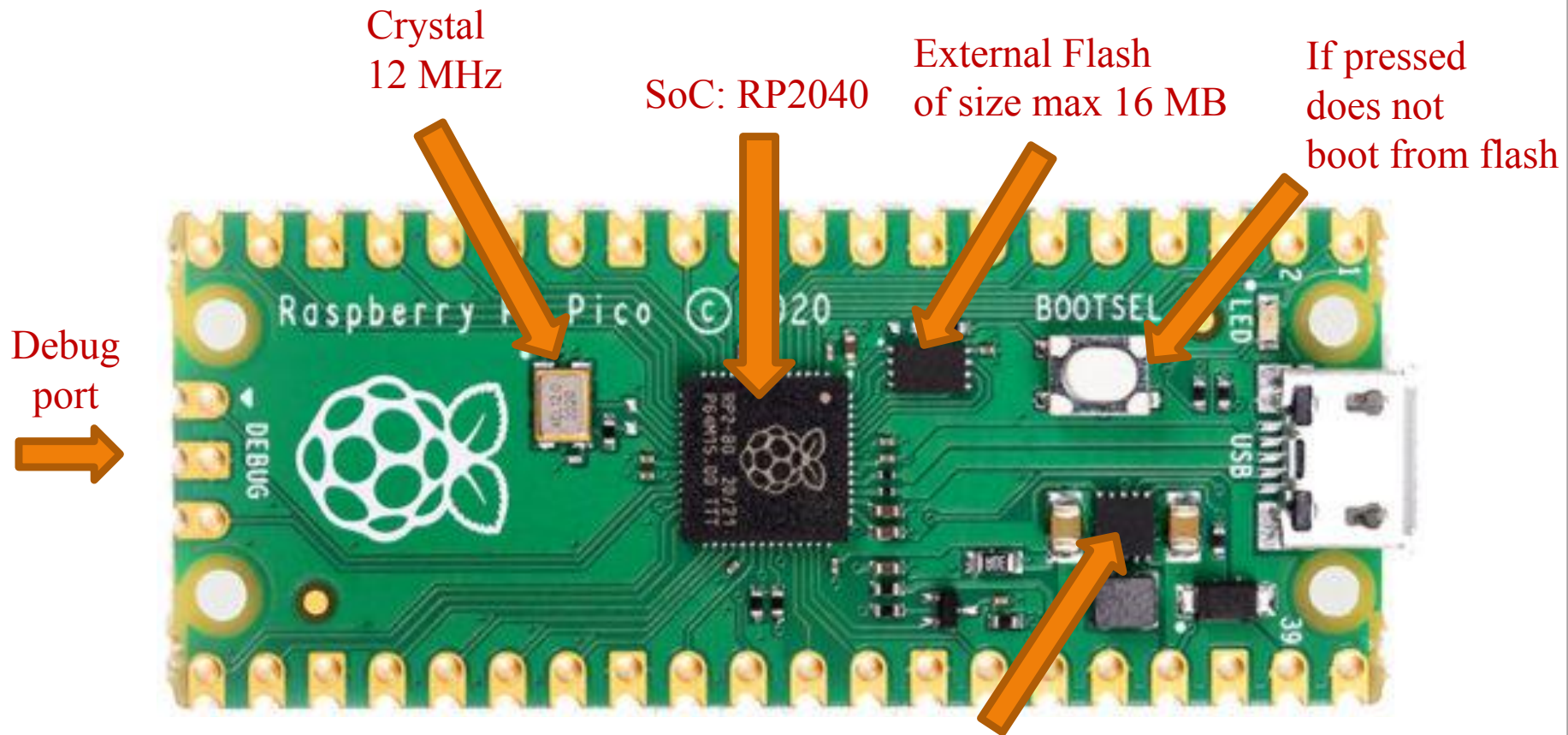
Front side



Back side



Raspberry Pi Pico (based on RP2040): Layout



SoC: System on a Chip

SMPS: Switched Mode Power Supply

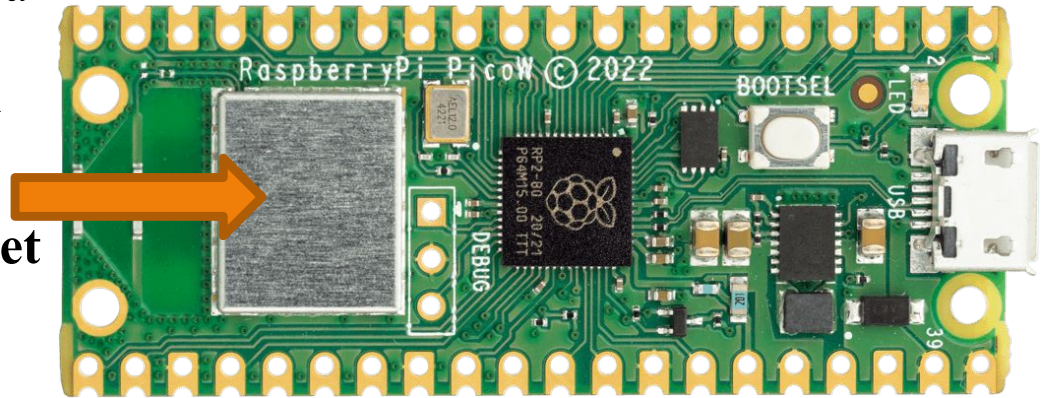
3.3 V Buck Boost
SMPS

Major Limitation:
No built-in WiFi module
in the Raspberry Pi Pico.

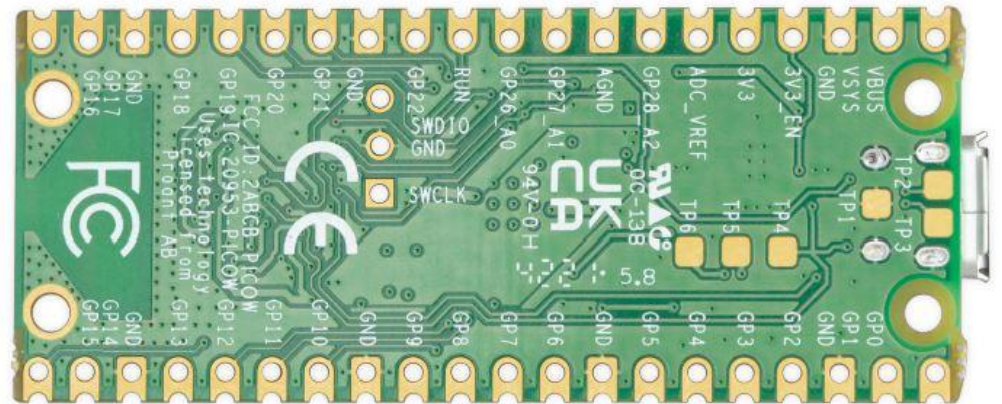
Raspberry Pi Pico W (based on RP2040)

- **Raspberry Pi Pico W** was released in **January 2022** with a retail price of **\$6**.
- A big change in the **Raspberry Pi Pico W** is the adding a **2.4 GHz 802.11b/g/n Wi-Fi chipset** and an on-board **antenna**.
- This is done without changing the **device's form factor (dimensions and pin layout)**, so we can swap a **Pico W** in for a **standard Pico** in any of your existing projects!!!
- The rest of the features and software support are almost unchanged.

Raspberry Pi Pico W (RP2040)
Front side



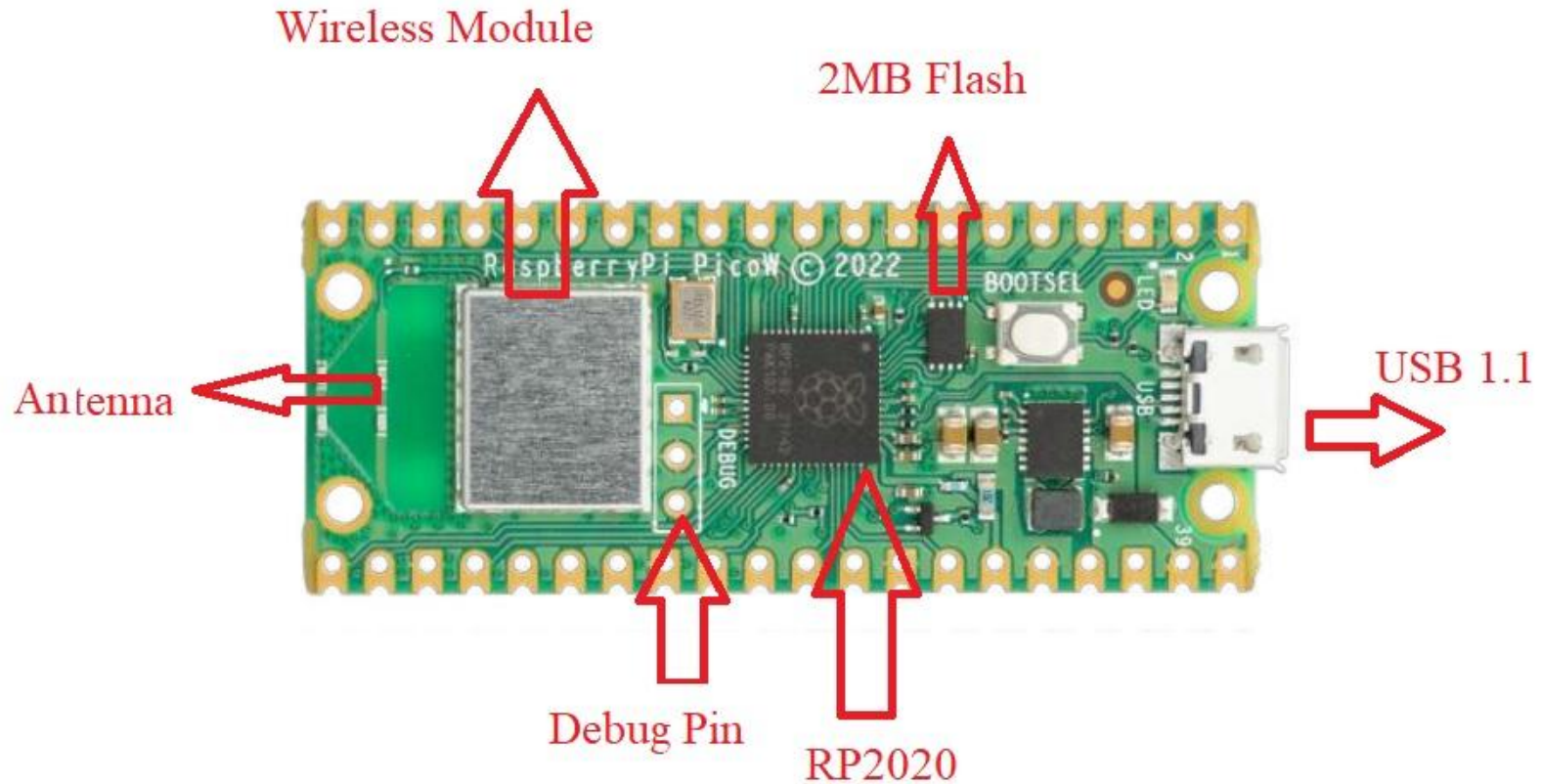
Back side



[Release news of Raspberry Pi Pico W](#)

Pin compatible with Raspberry Pi Pico

Raspberry Pi Pico W: Layout



Lecture 1.2.5: Summary

- Introduction to Raspberry Pi Family
 - Introduction
 - Raspberry Pi Pico
 - Raspberry Pi Pico W